WEEK 3 REPORT

THE CLEANING SPRINT

Project: Moneyball – IPL Player Valuation
Course: 24ADI204 – Data Science and Visualization
Event: Data-Viz Hack-A-Bit (2026–2027)
Team Number: 05

Team Member	Register Number
Divyesh D S	25BAD022
Kavi Prakash M	25BAD043
Mithuin P S	25BAD057
Madhan P M	25BAD503
Renganathan G M	25BAD087

Submitted as the Week 03 practical documentation for the team project.

1. Executive Summary

Week 03 focused on the cleaning sprint for the IPL Moneyball dataset. The work examined missing values, investigated whether selected missing performance statistics were genuinely undefined, treated selected categorical missing values, checked duplicate records and detected potential outliers using the IQR method and Z-scores. The notebook preserved the original dataset for a transparent before-and-after comparison.

Key results: 3,311 rows × 27 columns; missing cells reduced from 17,839 to 13,099 after labelling missing role, nationality and bowling style values as “Unknown”; zero duplicate rows; outliers detected but not automatically removed.

2. Objectives

• Inspect missing values and understand their possible meaning.
• Investigate missing batting and bowling statistics using participation indicators.
• Apply a defensible strategy to selected categorical missing values.
• Check duplicate records.
• Detect potential outliers using boxplots, IQR and Z-score.
• Compare the original and processed dataset without making unsupported changes.

3. Dataset Profile

The notebook loaded the merged IPL master dataset **IPL_Moneyball_Merged_2008_2026.csv**. A copy named **original_df** was retained before treatment so that the original data could be compared with the working dataset.

Baseline measure	Before cleaning
Rows	3,311
Columns	27
Missing cells	17,839
Duplicate rows	0

4. Missing-Value Analysis

The raw dataset contained 17,839 missing cells. The columns with the largest missing counts included:

Column	Missing values	Missing percentage
base_price_lakh	3,109	93.90%
role	2,418	73.03%
nationality	2,296	69.34%
bowling_avg	1,520	45.91%
final_price_lakh	1,480	44.70%
economy	1,130	34.13%
batting_avg	677	20.45%
strike_rate	383	11.57%

Column	Missing values	Missing percentage
bowl_style	26	0.79%

Auction-related columns status, team, auction_name and acquisition_type each contained 1,200 missing values. The notebook also identified 1,200 records in which all six selected auction-related fields were missing.

5. Missing-Value Investigation and Treatment

The notebook checked missing batting average, strike rate, bowling average and economy against participation indicators (dismissals, balls faced, wickets and balls bowled). The executed checks returned zero records for each of the four tested conditions, indicating no missing value in those statistics when the corresponding participation indicator was positive in the tested data.

Treatment applied: missing values in role (2,418), nationality (2,296) and bowl_style (26) were labelled **Unknown**. This handled 4,740 missing cells without guessing a player's category. Missing auction details and numerical performance statistics were retained where their meaning could not be safely inferred. Undefined averages should not be filled with zero or an overall mean/median because that would misrepresent player performance.

6. Duplicate Checking

The original dataset contained zero duplicate rows. The duplicate-removal check therefore removed no records, and the row count remained 3,311.

7. Graphs and Visual Analysis

The following representative graphs are taken from the executed Week 03 notebook evidence. They illustrate the missing-value pattern and potential outliers. The full notebook retains the remaining diagnostic plots.

7.1 Missing-Value Heatmap

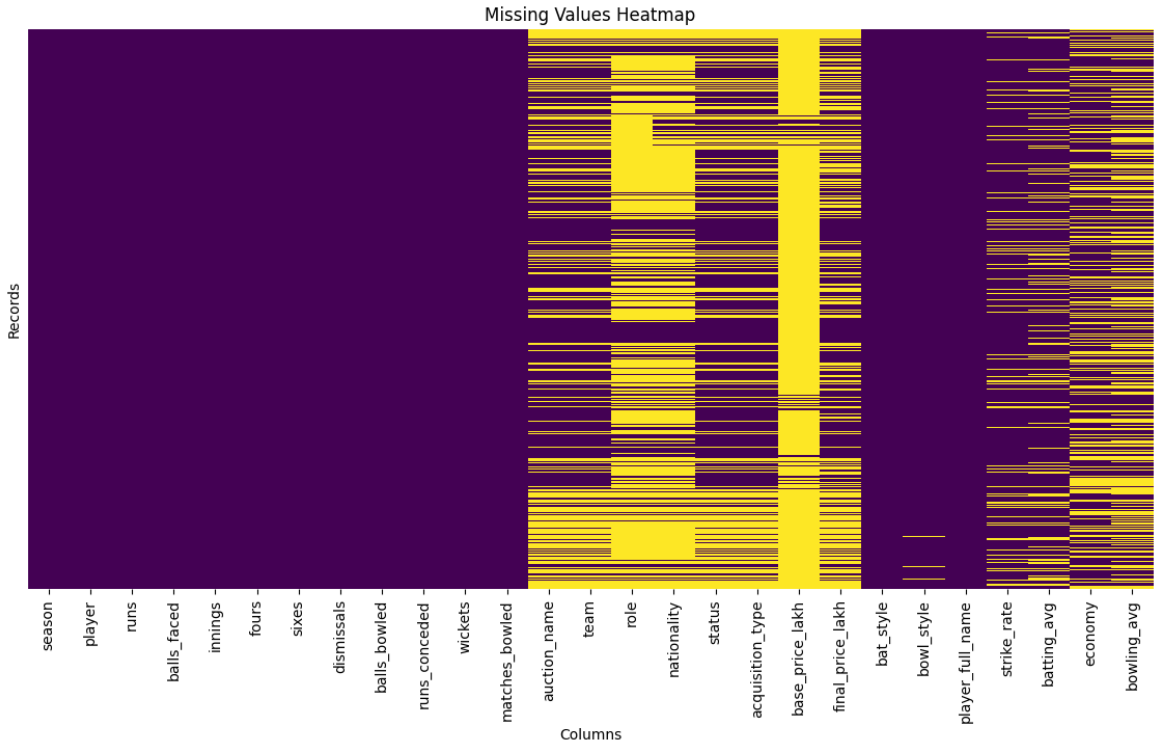


Figure 1. Missing-value heatmap showing the distribution of missing entries across dataset columns.

The heatmap highlights that missingness is concentrated in selected auction and player-profile fields rather than being evenly distributed across all columns.

7.2 Runs Boxplot

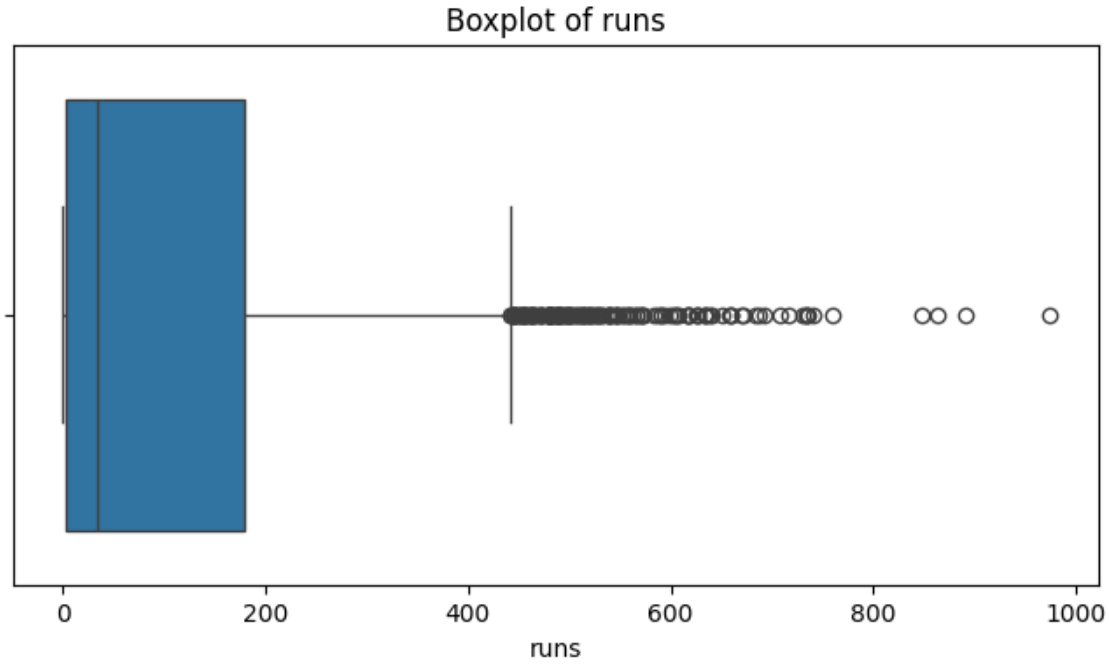


Figure 2. Boxplot of runs showing potential extreme observations identified by the IQR approach.

7.3 Final Auction Price Boxplot

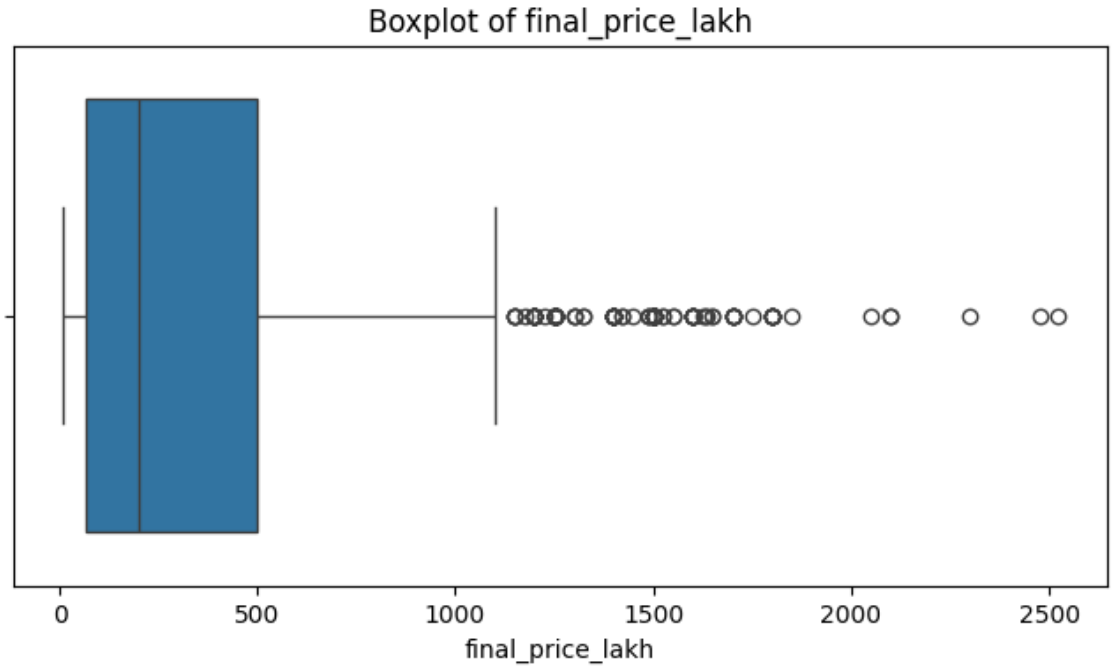


Figure 3. Boxplot of final_price_lakh showing potential high-value outliers.

The boxplots are diagnostic tools. A point outside the whiskers is a potential outlier, not automatically an incorrect record. IPL players can have genuinely exceptional performance values and auction prices.

8. Outlier Detection

IQR detection was applied to runs, strike_rate, batting_avg, wickets, economy and final_price_lakh. Z-score (absolute Z > 3) was calculated for runs, strike_rate, batting_avg, wickets and economy. final_price_lakh was not included in the Z-score calculation.

Column	IQR outliers	Z-score outliers
runs	156	46
strike_rate	182	20
batting_avg	39	25
wickets	203	44
economy	107	31
final_price_lakh	130	Not calculated

Outlier treatment decision: selected non-negative performance and price columns were checked for negative values; all reported negative-value counts were zero. Detected outliers were not automatically deleted or capped. Extreme IPL performances or auction prices may be genuine records. IQR and Z-score counts are column-level counts and should not be added as if they represented unique players.

9. Before-and-After Comparison

Measure	Before	After
Rows	3,311	3,311
Columns	27	27
Missing cells	17,839	13,099
Missing cells handled	—	4,740
Duplicate rows	0	0

The 4,740-cell reduction comes exactly from replacing missing role (2,418), nationality (2,296) and bowl_style (26) values with Unknown. The remaining 13,099 missing cells are documented, not silently removed or imputed. The unchanged row and column counts are expected because the selected treatment changed categorical values but did not delete records or columns.

10. Final Dataset and Output

The notebook saves the working dataset as **IPL_Moneyball_Cleaned_Week03.csv**. The executed final verification reported shape **(3,311, 27)**, **13,099** remaining missing cells and **zero** duplicate rows. The original dataset remains preserved separately as original_df.

11. Results and Interpretation

• Categorical missing-value treatment reduced the missing-cell count by 4,740.
• No duplicate rows were found, so no records were removed for duplication.
• IQR and Z-score methods identified potential outliers in several performance variables.
• Outliers were retained because extreme values may represent genuine player performance or auction outcomes.
• Missing auction details and numerical values were retained where the correct value could not be safely inferred.

12. Conclusion

Week 03 completed missing-value profiling, targeted categorical missing-value treatment, duplicate checking, boxplot visualization, IQR detection, Z-score detection and before-and-after verification. This is a targeted cleaning and outlier-review result, not a claim that every missing value or outlier has been eliminated. The retained missing values and valid extreme observations must be considered during Week 04 EDA.

13. Team Members

The project team consists of the following members:

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Divyesh D S	25BAD022
Kavi Prakash M	25BAD043
Mithuin P S	25BAD057
Madhan P M	25BAD503
Renganathan G M	25BAD087

Note: Individual task allocations were not provided, so specific contributions are not assigned in this report.

14. Project Files and References

- Week 03 notebook: Week03_Cleaning_Sprint_Final.ipynb.
- Cleaned output: IPL_Moneyball_Cleaned_Week03.csv.
- Original master dataset: IPL_Moneyball_Merged_2008_2026.csv (team-provided merged dataset).
- Data-Viz Hack-A-Bit (2026–2027), course 24ADI204 – Week 03 activity requirements.

Limitation: The notebook does not establish the original provenance or collection method for every missing auction/profile field. Therefore, no unsupported assumptions about the cause of every missing value are made.